



FEE247 - DESENVOLVIMENTO DE SOLUÇÕES IOT COM LORA E LORAWAN

Módulo 6 - LoRaWAN

Tópico 6.2 - Gateway Monocanal LoRaWAN - Firmware e declaração na TTN

FEE-247 - DESENVOLVIMENTO DE SOLUÇÕES IOT COM LORA E LORAWAN

Aula 16-05-2026 - Gateway Monocanal PKLoRaWAN

TTN Map

<https://www.thethingsnetwork.org/map>

Heat Map TTN Mapper

<https://ttnmapper.org/heatmap/>

Site TTN

<https://www.thethingsnetwork.org/>

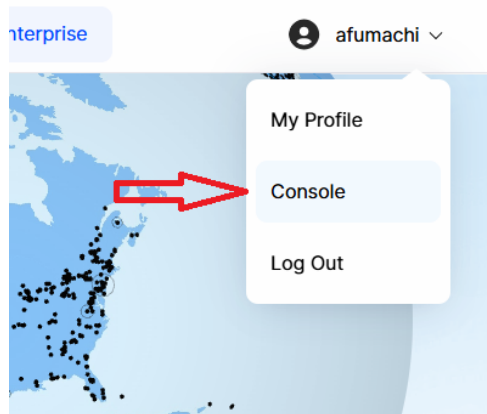
Onde criar um ID na TTN

<https://www.thethingsnetwork.org/login/>

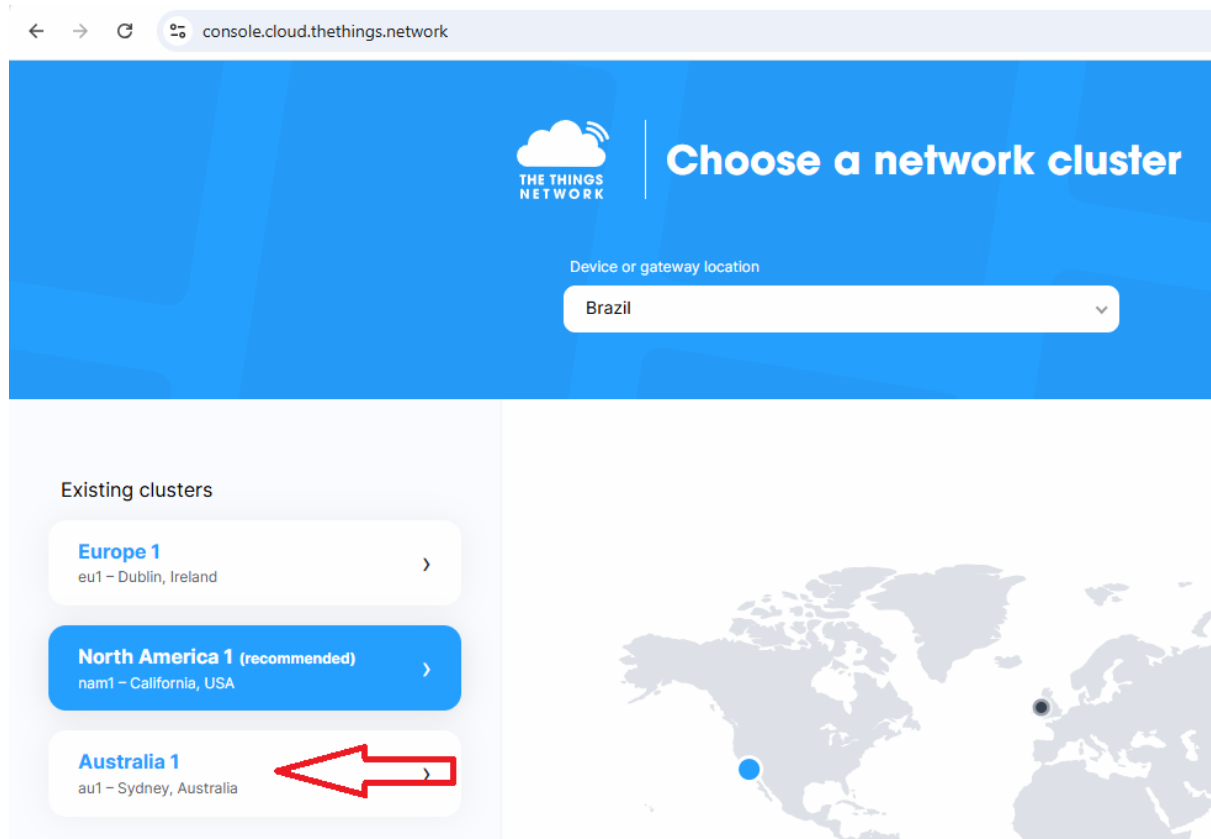


Realizar Login , ou criar Login

Acessar o Console



Acessar o Cluster AU1 => Austrália 1



Realizar o Login no SANDBOX da TTN, caso necessário

Configuração Firmware Gateway Monocanal

Arquivo



loraModem.h

Diagrama de Pinos do ESP32 ↔ RFM95 - _PIN_OUT==4 compatível com PKLoRaWAN

```

260 #elif _PIN_OUT==4
261 // -----
262 // For ESP32/TTGO based board. //placa RF95
263 // SCK == GPIO5/ PIN5
264 // SS == GPIO18/ PIN18 CS
265 // MISO == GPIO19/ PIN19
266 // MOSI == GPIO27/ PIN27
267 // RST == GPIO14/ PIN14
268 struct pins {
269     uint8_t dio0=26; // GPIO26/ Dio0 used for one frequency and one SF
270     uint8_t dio1=35; // GPIO35/ Used for CAD, may or not be shared with DI00
271     uint8_t dio2=34; // GPIO34/ Used for frequency hopping, don't care
272     uint8_t ss=18; // GPIO18/ CS. Select pin connected to GPIO18
273     uint8_t rst=14; // GPIO14/ D3. Reset pin not used
274 } pins;
275 #define SCK 5
276 #define MISO 19
277 #define MOSI 27
278 #define RST 14
279 #define SS 18

```

Frequência selecionada no arquivo principal

```

127 #elif defined(AU915_928)
128 // Australian plan or TTN/Lora frequencies
129 vector freqs [] = {
130     { 916800000, 125, 7, 10, 916800000, 125, 7, 12}, // Channel 0, 916.8 MHz primary
131     { 917000000, 125, 7, 10, 917000000, 125, 7, 12}, // Channel 1, 917.0 MHz mandatory
132     { 917200000, 125, 7, 10, 917200000, 125, 7, 12}, // Channel 2, 917.2 MHz mandatory
133     { 917400000, 125, 7, 10, 917400000, 125, 7, 12}, // Channel 3, 917.4 MHz Optional
134     { 917600000, 125, 7, 10, 917600000, 125, 7, 12}, // Channel 4, 917.6 MHz Optional
135     { 917800000, 125, 7, 10, 917800000, 125, 7, 12}, // Channel 5, 917.8 MHz Optional
136     { 918000000, 125, 7, 10, 918000000, 125, 7, 12}, // Channel 6, 918.0 MHz Optional
137     { 918200000, 125, 7, 10, 918200000, 125, 7, 12}, // Channel 7, 918.2 MHz Optional
138     { 917500000, 500, 8, 8, 0, 0, 0, 0}, // Channel 8, 917.5 SF8BW500 MHz Optional Uplink
139 };
140

```

Arquivo



configNode.h

Rede Wi-Fi e Declaração do Nome do Gateway (Gateway_ID) e localização

```

97 // Please fill in at least ONE valid SSID and password from your own WiFi network
98 // below. This is needed to get the gateway working
99 // ADICIONE AQUI AS SUAS REDES WI-FI
100 wpa_wpa[] = {
101     {"", ""}, // Reserved for WiFi Manager.
102     {"SEU_WIFI_SSID", "SEU_WIFI_PASSWORD"}
103 };
104 // If you have a second back-end server defined such as Semtech or loriot.io
105 // you can define _THINGPORT and _THINGSERVER with your own value.
106 // If not, make sure that you do not define these, which will save CPU time
107 // Port is UDP port in this program
108 //
109 // Default for testing: Switched off
110 // #define _THINGSERVER "" // Server URL of the LoRa-udp.js handler
111 // #define _THINGPORT 1700 // Port 1700 is old compatibility
112
113 // Define the name of the accesspoint if the gateway is in accesspoint mode (is
114 // getting WiFi SSID and password using WiFiManager).
115 // If you do not need them, comment out.
116 // #define AP_NAME ""
117 // #define AP_PASSWD ""
118
119 // Gateway Ident definitions. Where is the gateway located?
120 #define _DESCRIPTION "SEU_GATEWAY_ID" // Name of the gateway
121 #define _EMAIL "SEU_E-MAIL" // Owner
122 #define _PLATFORM "ESP32" // Platform
123 #define _LAT -22.820923 // Latitude
124 #define _LON -47.066795 // Longitude
125 #define _ALT 625 // Altitude
126

```

Arquivo

configGWay.h

Frequência do Gateway Monocanal

```

78 // Define the frequency band the gateway will listen on. Valid options are
79 // EU863_870 Europe
80 // US902_928 North America
81 // AU915_928 Australia
82 // AU915_928LA Brazil
83 // CN470_510 China
84 // IN865_867 India
85 // CN779-787 (Not Used)
86 // EU433 Europe
87 // KR920 South Korea
88 // AS923 (Not Used)
89 // You can find the definitions in "loraModem.h" and frequencies in
90 // See https://www.thethingsnetwork.org/docs/lorawan/frequency-plans.html
91 #define AU915_928 1
92

```

Definição do Diagrama de Pinos _PIN_OUT==4 compatível com PKLoRaWAN

```

160 // We support a few pin-out configurations out-of-the-box: HALLARD, COMPRESULT and
161 // Heltec/TTGO ESP32.
162 // If you use one of these, just set the parameter to the right value.
163 // If your pin definitions are different, update the loraModem.h file to reflect the
164 // hardware settings.
165 // 1: HALLARD
166 // 2: COMRESULT pin out
167 // 3: ESP32, Wemos pin out (Not used)
168 // 4: ESP32, Heltec and TTGO pin out (should work for Heltec, 433 and Oled too).
169 // 5: Other, define your own in loraModem.h (does not include GPS Code)
170 #if !defined _PIN_OUT
171 #   define _PIN_OUT 4
172 #endif
173

```

Desabilita Display Oled

```

215 // Define if Oled Display is connected to I2C bus. Note that defining an Oled display does not
216 // impact performance negatively, certainly if no Oled is connected. Wrong Oled will not show
217 // sensible results on the Oled display
218 // _OLED==0;    No Oled display connected
219 // _OLED==1;    0.9" Oled Screen based on SSD1306
220 // _OLED==2;    1.3" Oled screens for Wemos, 128x64 SH1106
221 #if !defined _OLED
222 #   define _OLED 0 // 2
223 #endif
224

```

Arquivo

_utils.ino

Linha 475, substituir o tipo da variável de “byte” para “uint8_t”

```

473 const int NTP_PACKET_SIZE = 48; // Fixed size of NTP record
474 //byte packetBuffer[NTP_PACKET_SIZE];
475 uint8_t packetBuffer[NTP_PACKET_SIZE];
476 memset(packetBuffer, 0, NTP_PACKET_SIZE); // Set buffer contents to zero
477

```

Arquivo ESP-sc-gway.ino => definição do Diagrama de Pinos

```

26
27 #if defined (ARDUINO_ARCH_ESP32) || defined(ESP32)
28 #   define ESP32_ARCH 1
29 #   ifndef _PIN_OUT
30 #       define _PIN_OUT 4 // For ESP32 pin-out 4 is default //colocar a placa aqui
31 #   endif
32 #elif defined(ARDUINO_ARCH_ESP8266)
33 //
34 #else
35 #   error "Architecture unknown and not supported"
36 #endif
37

```

Garantir a função `WiFi.setAutoReconnect(true);`

Caso estiver com a função `WiFi.setAutoConnect(true);`



Modificar para setAutoReconnect

```

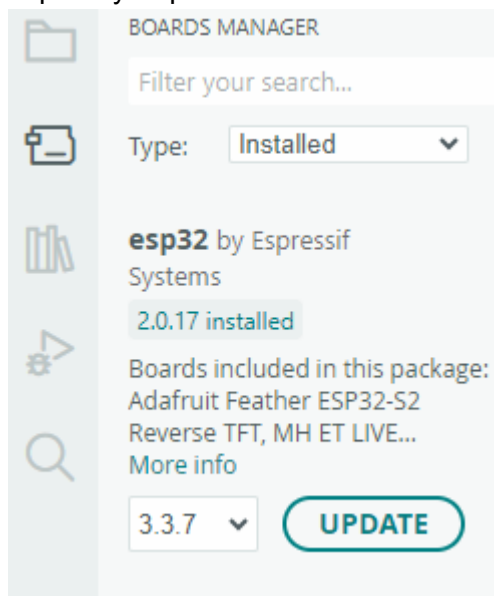
420
421     msg_oLED("WIFI_STA");
422     WiFi.mode(WIFI_STA);                                     // WiFi settings for connections
423     WiFi.setAutoReconnect(true);
424     WiFi.macAddress(MAC_array);
425     sprintf(MAC_char, "%02x:%02x:%02x:%02x:%02x:%02x",
426             MAC_array[0], MAC_array[1], MAC_array[2], MAC_array[3], MAC_array[4], MAC_array[5]);
427
428 #   if _MONITOR >= 1
429     mPrint("MAC: " + String(MAC_char) + ", len=" + String(strlen(MAC_char)) );
430 #   endif // _MONITOR
431

```

Arduino => Board Manager

[GitHub - espressif/arduino-esp32: Arduino core for the ESP32 · GitHub](https://github.com/espressif/arduino-esp32)

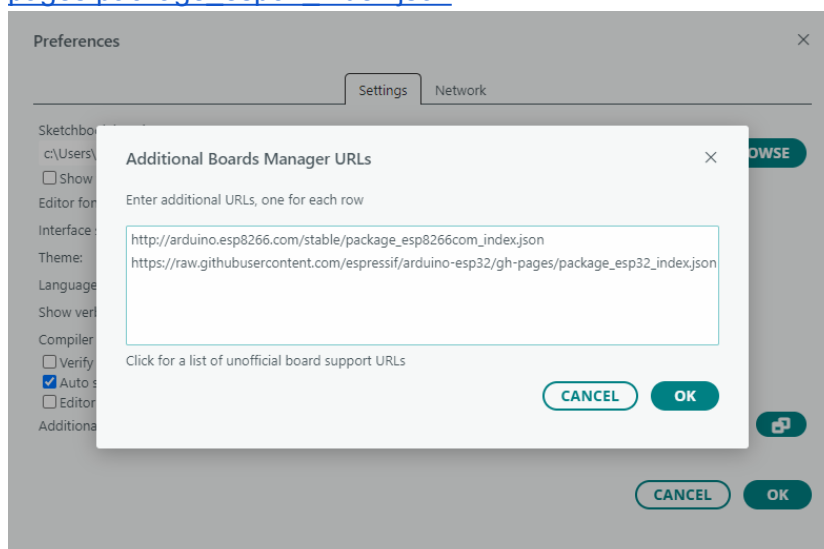
esp32 by Espressif => Vs. 3.3.7



Arduino => File => Preferences...

Additional Boards Manager URLs:

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json



Bibliotecas e Versões necessárias:

[Adafruit_BusIO](#) => Vs 1.17.4

Adafruit BusIO by
Adafruit

1.17.4 installed

This is a library for abstracting away UART, I2C and SPI interfacing This is a library for...

[More info](#)

1.17.4 ▾

REMOVE

[Adafruit-GFX-Library](#) => Vs 1.12.4

Adafruit GFX Library
by Adafruit

1.12.4 installed

Adafruit GFX graphics core library, this is the 'core' class that all our other graphics...

[More info](#)

1.12.5 ▾

UPDATE

[Adafruit_NeoPixel](#) => Vs 1.9.0

Adafruit NeoPixel by
Adafruit

1.9.0 installed

Arduino library for controlling single-wire-based LED pixels and strip. Arduino library for...

[More info](#)

1.15.4 ▾

UPDATE

[Adafruit_SSD1306](#) => Vs 2.5.16

Adafruit SSD1306 by
Adafruit
2.5.16 installed
SSD1306 oled driver library for
monochrome 128x64 and
128x32 displays SSD1306 oled...
[More info](#)
2.5.16 ▼ **REMOVE**

[ArduinoJson: Efficient JSON serialization for embedded C++](#)

Arduinojson => Vs 7.4.3

ArduinoJson by Benoit
Blanchon...
6.14.0 installed
A simple and efficient JSON
library for embedded C++. ★
7124 stars on GitHub Support...
[More info](#)
7.4.3 ▼ **UPDATE**

[GitHub - succulent/esp32-http-update: Clone of](#)

<https://github.com/esp8266/Arduino/tree/master/libraries/ESP8266httpUpdate> for ESP32 ·
[GitHub](#)

ESP32httpUpdate => Vs 2.1.145

ESP32httpUpdate by
Matej Sychra
2.1.145 installed
Http Update for ESP32 Quick
clone of Arduino
ESP8266httpUpdate for ESP32...
[More info](#)
2.1.145 ▼ **REMOVE**

[GitHub - ThingPulse/esp8266-oled-ssd1306: Driver for the SSD1306 and SH1106 based 128x64, 128x32, 64x48 pixel OLED display running on ESP8266/ESP32 · GitHub](#)

ESP8266 and ESP32 OLED driver for... => Vs. 4.4.1

**ESP8266 and ESP32
OLED driver for...**

4.4.1 installed

I2C display driver for SSD1306
OLED displays connected to
ESP8266, ESP32, Mbed-OS The...

[More info](#)

4.6.2 ▾ **UPDATE**

HotButton => Vs. 0.1.1

HotButton by Rop
Gonggrijp...

0.1.1 installed

Button library for Arduino. The
usual, plus "if
(button.event(LONG, SHORT,...

[More info](#)

0.1.1 ▾ **REMOVE**

[GitHub - mcci-catena/arduino-lmic: LoraWAN-MAC-in-C library, adapted to run under the Arduino environment · GitHub](#)

MCCI LoRaWAN LMIC library => Vs. 4.1.1

**MCCI LoRaWAN LMIC
library** by IBM, Matthijs...

4.1.1 installed

Arduino port of the LMIC
(LoraWAN-MAC-in-C) framework
provided by IBM. Supports...

[More info](#)

6.0.0 ▾ **UPDATE**

[GitHub - ropg/LoRaWAN_ESP32: ESP32 persistence \(deep sleep\) code for RadioLib LoRaWAN. Can also manage provisioning data · GitHub](#)

LoRaWAN_ESP32 => Vs. 1.2.0

LoRaWAN_ESP32 by Rop Gonggrijp...

1.2.0 installed

ESP32 'persist' object for using deep sleep in combination with RadioLib LoRaWAN. Also allow...

[More info](#)

1.2.0 ▼ REMOVE

[GitHub - janelia-arduino/Streaming: Streaming C++-style Output with Operator << · GitHub](#)

Streaming => Vs. 5.0.0

Streaming by Mikal Hart <mikal@arduiniana.org>

5.0.0 installed

Streaming C++-style Output with Operator << Like this project? Please star it on...

[More info](#)

6.2.0 ▼ UPDATE

[Arduino Playground - Time](#)

Time => Vs. 1.6

Time by Michael Margolis

1.6 installed

Timekeeping functionality for Arduino Date and Time functions, with provisions to...

[More info](#)

[GitHub - tzapu/WiFiManager: ESP8266 WiFi Connection manager with web captive portal · GitHub](#)

WiFiManager => Vs 2.0.16-rc-2

WiFiManager by tzapu

2.0.16-rc.2 installed

WiFi Configuration manager with web configuration portal for Espressif ESPx boards, by...

[More info](#)

2.0.17 ▼ UPDATE

[GitHub - Densaugeo/base64_arduino: Base64 encoder/decoder for arduino repo · GitHub](#)

base64 => Vs. 1.0.0

base64 by Densaugeo
<use@git.hub>
1.0.0 installed
Base64 encoder/decoder for
arduino repo Uses common
web conventions - '+' for 62, '/'...
[More info](#)

1.3.0 ▾ **UPDATE**